

CAD-CBA: A Class-Aware Distilled CNN–BiLSTM for Multi-Objective IoT Intrusion Detection with Operation-Matched CUDA Acceleration

Document type: Camera-ready **local-complete** manuscript (PI venue polish pass) **Method freeze:** CAD-CBA-v1 · **Option A** CUDA (per-block / operation-matched only) **Claims package:** 59 protocol claims · PYTHONPATH=. python3 scripts/verify_claims.py must stay green **Champion weights:** model/best_model_botiot_twostage.pth · md5 80a90f7cc210276300eaa90173a5a385 **Authority for numbers:** docs/execution_plan/CLAIMS_REGISTRY.md + on-disk benchmarks/results/ **Open ops (not claimed):** DICC multi-GPU / multi-day cells remain **TBD** — never invent

Field	Value
Title (locked T1)	CAD-CBA: A Class-Aware Distilled CNN–BiLSTM for Multi-Objective IoT Intrusion Detection with Operation-Matched CUDA Acceleration
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Keywords	IoT intrusion detection; class imbalance; knowledge distillation; multi-objective evaluation; CUDA acceleration; BoT-IoT; protocol-fair baselines

***Scope of this draft.** Local science and local systems evidence are complete. Multi-GPU portability is **not** claimed until a dedicated DICC SUCCESS tree exists. All load-bearing numbers trace to sealed JSON on this machine.*

Abstract

IoT intrusion detection must handle extreme class imbalance and low-latency edge constraints. Tree ensembles often top pure accuracy on static tabular flows, while deep models remain preferred for GPU deployment, incremental adaptation, and systems co-design. Unfair protocols and framework-only acceleration claims, however, obscure real trade-offs: prior work often siloes detection quality from deployment metrics, tunes without a sealed test, or reports single-shot laptop latencies as multi-platform facts. Full-pipeline “Custom CUDA versus full model” comparisons are also invalid when kernels implement only operation-matched blocks (**Option A**).

We freeze **CAD-CBA-v1**: a V3 CNN–BiLSTM–Attention student distilled from an **ensemble** of tree teachers (mean RF+XGB+LGBM soft labels, $\alpha=0.6$, $T=10$); **focal** loss ($\gamma \approx 1.92$); Optuna train hyperparameters (config/hpo_best.yaml); **shuffle** sampling; and **argmax** decode. Evaluation uses protocol botiot_v1 with fair classical and neural baselines, a full ablation ladder, multi-objective Pareto analysis, local multi-session systems ranges under Option A CUDA, structured evidence with dispatch timing, and a ToN recipe-transfer honesty check.

On BoT-IoT **sealed multi-seed TEST** ($n=5$, freeze path A), CAD-CBA-v1 reaches macro-F1 **0.9780±0.0033**, min-class **0.9292**, and Theft **1.0000**. Protocol classical pure-F1 remains led by LightGBM (**0.9818** val) with RF at **0.9778** val; the published RF **0.9864** is a **different pipeline** and is shown only as a dual bar. The package ladder tops at A7 **0.9699** (seed42), while attention+CE alone **hurts** versus plain CNN–BiLSTM (A4 **0.7378** ■ A3 **0.9493**). Multi-objective ranking places G6 MLP first on an a priori composite (**0.9056 @ 4.33** μ s, F1 0.9285). Local systems (WP6b, $n=5$ sessions, RTX 3050) report energy **0.920–0.943** mJ/flow (mean **0.933**, CV **1.05%**), PT@256 **24.15–25.68** μ s/sample (mean **24.90**, CV **2.22%**), CUDA derived pipeline **565.2–570.3** μ s (Option A), and peak alloc **322.2** MiB. Explainability is scoped: dispatch p99 **16.60** μ s, rank correlation **0.9636**, faithfulness mass **0.5109**, free-form LLM feature-mention **0.333** — **no** full explainable title claim. On ToN (13-feat), test **0.8110** lags same-split RF **0.9393** (honest gap).

Under a sealed, protocol-fair evaluation, CAD-CBA-v1 delivers **near-RF multi-seed test detection** with **strong minority (Theft) recognition**. The primary scientific claim is a **valid accuracy–efficiency multi-objective story** with **operation-matched CUDA** and **local multi-session latency/energy ranges** — not pure-F1 supremacy over LightGBM, and not multi-GPU portability until DICC completes.

1. Introduction

1.1 Motivation

Edge and campus IoT deployments need detectors that (i) remain robust under severe class imbalance, (ii) support GPU inference with predictable latency and energy, and (iii) admit honest comparison against strong classical baselines on the **same** data protocol. Much of the literature optimises one axis (pure F1, or single-shot latency) and then generalises beyond the measured hardware and training recipe. Professor feedback on our interim report emphasised multi-objective contribution, protocol-fair baselines, sealed-test discipline, and systems claims that do not over-reach the measured

platform.

1.2 Research questions

RQ	Question	Locked answer (this paper)
RQ1	Does CAD-CBA improve detection under protocol-fair evaluation?	Sealed multi-seed test macro-F1 0.9780±0.0033 ≈ protocol RF val 0.9778 ; does not beat protocol LGBM val 0.9818 or published RF 0.9864 (other pipeline). Detection alone is not the sole headline.
RQ2	Does it improve minority recognition?	Yes on protocol test : Theft mean 1.0 , min-cls mean 0.9292 (n=5).
RQ3	Does it reduce latency or memory on the measured GPU?	Yes on local RTX 3050 : PT@256 24.15–25.68 μs/sample; energy 0.920–0.943 mJ/flow; peak alloc 322.2 MiB; multi-obj composite G6 0.9056 @4.33 μs.
RQ4	Are systems results valid across GPUs / multi-day?	Not yet claimable — local ranges ≠ multi-day V100S/A100 (DICC open).
RQ5	Does explainability add measurable value?	Dispatch + structured evidence yes; free-form LLM quality no (full explainable claim dropped).

1.3 Contributions

- **CAD-CBA-v1 method package** fully frozen and evaluated: ensemble KD + focal + Optuna train HPs + V3 backbone under protocol `botiot_v1`, with sealed multi-seed BoT **test**.
- **Protocol-fair classical and neural baselines** on the same splits, with dual-bar honesty versus published RF.
- **Ablation and negative results** showing package-level credit (A7) and rejecting attention-alone, multi-scale CNN, gated fusion, SupCon, ASL, neural-teacher KD, and stratified batching.
- **Multi-objective Pareto** analysis and **local multi-session** latency/energy/VRAM ranges under Option A CUDA discipline.
- **Scoped XAI**: low-overhead dispatch + structured evidence; no full “LLM-explainable IDS” title claim.
- **ToN recipe-transfer honesty** on 13-feat processed data with same-split RF gap disclosed.

1.4 Non-claims (explicit)

- We do **not** claim pure-F1 supremacy over protocol LightGBM.
- We do **not** claim full Custom-CUDA versus full V3 PyTorch **parity** (Option A).
- We do **not** claim multi-GPU / multi-day portability until DICC artifacts exist.
- We do **not** claim full free-form LLM explainability (feature-mention **0.333**).

2. Related work

2.1 Deep and hybrid IDS for IoT / tabular flows

CNN, LSTM/BiLSTM, and hybrid CNN–RNN stacks are widely applied to flow and packet features. Many report high accuracy on public datasets but omit sealed-test discipline, protocol-matched classical baselines, or multi-seed stability. Transformer-style temporal models are sometimes proposed as drop-in upgrades; under our equal-budget protocol baseline they underperform (G12 val macro-F1 **0.5808** ■ G11 **0.9493**).

2.2 Class imbalance in intrusion detection

Focal loss, class-balanced losses, oversampling (e.g. SMOTE), and threshold search are common. Our four-way loss compare under protocol fine-tuning finds **focal** best (val **0.9780**); class-balanced focal (**0.9121**) and logit adjustment (**0.9225**) hurt macro-F1. Validation threshold search does not beat **argmax**. Stage-A SMOTE remains a KD-path tool; Stage-B fine-tune uses real train only.

2.3 Knowledge distillation and teacher ensembles

Distilling tree ensembles into compact neural students is attractive for deployment. We compare RF, XGB, LGBM, none, and **ensemble** soft-label teachers under identical Stage-A KD; the ensemble student val **0.9401** wins. A strong neural teacher (G11) yields a weaker student (**0.8513**).

2.4 GPU acceleration and systems claims

Framework compilers (TensorRT, ORT, `torch.compile`) and hand-written CUDA kernels both appear in IDS systems papers. Invalid comparisons arise when custom kernels cover only operation-matched blocks but are reported as

full-model speedups. We adopt **Option A**: report per-block / derived pipeline CUDA latency and full V3 PyTorch absolute latency **separately**, never as parity.

2.5 Explainable IDS and LLMs

Recent LLM-assisted IDS work often emphasises narrative quality without faithfulness/consistency metrics, or conflates prompt-dispatch cost with generation latency. We measure occlusion faithfulness (**0.5109** top-3 mass), rank consistency (**0.9636**), dispatch p99 (**16.60** μ s), generation mean (**~7400** ms), and free-form feature-mention (**0.333**), then **drop** a full explainable title claim while retaining structured evidence + dispatch.

2.6 Positioning gap table

Gap in prior practice	Our response	Evidence
Test used during selection	Sealed test until freeze lock (B14)	FINAL_CONFIG_FREEZE_CARD.md, sealed_test/
Classical baselines on other pipelines	Protocol-fair RF/XGB/LGBM/SVM/LR	baselines_classical/
Single-shot laptop latency as “the” number	Multi-session ranges n=5	wp6b_local_ranges/
Full-pipeline CUDA vs full PT false parity	Option A language + fidelity	numerical_fidelity.json
Architecture novelty without ablation	A1–A7 + C* negatives	ablation_ladder/, cstar_bounded/
Full LLM-XAI claim without metrics	Drop full claim; keep structured	xai/summary.json
Cross-GPU claim from one laptop	Explicit TBD cells	DICC BLOCKED

3. Method: CAD-CBA-v1

3.1 Architecture (V3)

Input: 10 BoT-IoT flow features $\rightarrow$ linear projection to 64 $\rightarrow$ reshape **[2, 32]** $\rightarrow$ two 1D-CNN layers (64 $\rightarrow$ 128, kernel 3, BN, ReLU, pool) $\rightarrow$ BiLSTM (128 $\rightarrow$ 64, bidirectional) $\rightarrow$ multi-head attention (4 heads) $\rightarrow$ dense 64 + dropout $\rightarrow$ 5-class softmax. Trainable parameters of the production package path: **530,181** (Pareto/systems tables).

CAD-CBA-v1 architecture (V3 CNN-BiLSTM-Attention)
dims: reshape [2,32] · CNN 64 $\rightarrow$ 128 k=3 · BiLSTM 128 $\rightarrow$ 64 · MHA heads=4 · head 64 · 5 classes

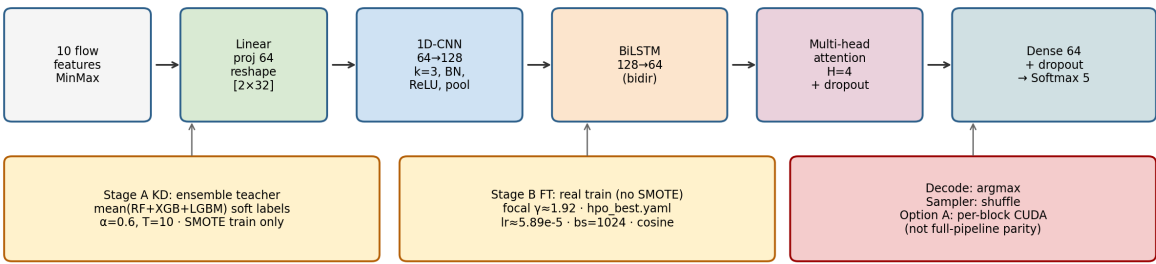


Figure 1. CAD-CBA-v1 architecture and two-stage training (dims from config/config.yaml; train hyperparameters from config/hpo_best.yaml).

3.2 Training recipe (frozen)

Stage	Role	Key settings
Stage A KD	Distill student from soft labels	Teacher = mean(RF, XGB, LGBM) probs; $\alpha=0.6$, $T=10$; SMOTE on train only; val for selection
Stage B FT	Real-data fine-tune	No SMOTE ; focal $\gamma=1.9166$; $lr=5.893e-5$; batch 1024 ; dropout ≈ 0.148 ; attn dropout ≈ 0.214 ; weight decay $\approx 1.92e-4$; cosine schedule; shuffle sampler
Decode	Inference decision	argmax (val threshold search: no macro/min lift)

Init path for sealed multi-seed test (**path A**): model/best_model_botiot_distill_a0.6_T10.0_focal2.pth + Stage-B FT with hpo_best.yaml, seeds 42–46.

3.3 Production champion (systems / XAI path)

Production weights used for systems ranges, the historical energy table, and the XAI suite:

- Path: model/best_model_botiot_twostage.pth
- md5: 80a90f7cc210276300eaa90173a5a385
- Never overwritten without BACKUP and explicit approval.

3.4 What is *not* in the package

Multi-scale CNN (C4 **0.9167**), gated fusion (C5 **0.9132**), SupCon (C7 **0.7732**), asymmetric loss (C8 **0.8012**), MC-dropout selective decoding (no high-coverage lift), neural teacher KD (E6 **0.8513**), full architecture Optuna (B2–B4 plateau reject), class-balanced weights on neural focal, and stratified inv-freq batching (D6 **0.9209** ■ shuffle **0.9791**).

4. Experimental protocol

4.1 Dataset and splits (BoT-IoT)

Protocol ID: botiot_v1. Features: 10 numerical flow features (freeze list in scripts/protocol/botiot.py). Classes: DDoS, DoS, Normal, Reconnaissance, Theft. Official test CSV is **sealed** for all model selection (HPO, thresholds, early stopping). Stage A uses stratified val + SMOTE on train only; Stage B uses stratified val + **no** SMOTE.

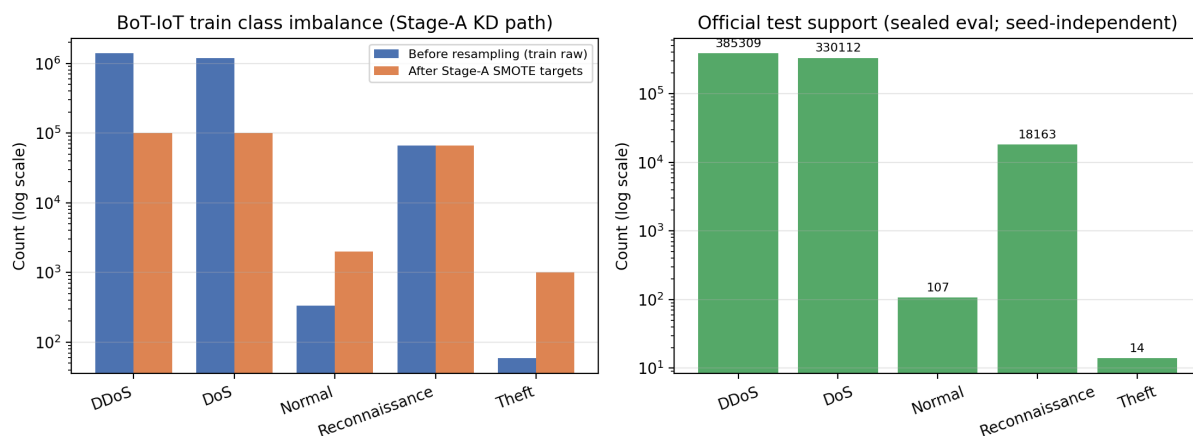


Figure 2. Extreme train imbalance and official test support (log scale). Theft test support is only **14** — minority metrics are reported with that honesty.

4.2 Metrics

Primary: **macro-F1**. Always log: balanced accuracy, weighted F1, min per-class F1, per-class F1 (especially Theft), and confusion matrices. Systems: μ s/sample (multi-session), mJ/flow, peak allocated VRAM, n_params, Option A CUDA block/pipeline μ s.

4.3 Seeds and hardware

- Multiruns / sealed test seeds: **42–46** (n=5).
- Local systems: NVIDIA GeForce **RTX 3050 6GB** Laptop GPU; WP6b **5** sessions; warm-up **50** sync forwards.
- DICC V100S/A100 multi-day: **not** in this draft's numeric tables.

4.4 Baselines (same protocol)

Classical: LR, LinearSVC, RF, XGB, LGBM (G5 balanced multiclass). **Neural (equal fixed CE budget, seed42):** MLP, 1D-CNN, LSTM, BiLSTM, CNN-LSTM, CNN-BiLSTM, lightweight temporal transformer. **Published RF 0.9864:** dual bar only (different pipeline) — never mixed as protocol-fair.

5. Results

5.1 Overall detection — sealed multi-seed TEST (B14)

Table 1. Sealed multi-seed BoT TEST (CAD-CBA-v1 path A; seeds 42–46). Source: benchmarks/results/sealed_test/summary.json.

Seed	val macro-F1	test macro-F1	test min-cls	test Theft
42	0.9791	0.9787	0.9333	1.0000
43	0.9587	0.9798	0.9369	1.0000
44	0.9797	0.9798	0.9375	1.0000
45	0.9787	0.9722	0.9014	1.0000
46	0.9483	0.9796	0.9369	1.0000
Mean±std	0.9689±0.0145	0.9780±0.0033	0.9292	1.0000

Honesty notes. Seed 46 has weaker **val** (0.9483) but strong **test** (0.9796) — both are reported. Do not replace the test mean with the val multirun **0.9714±0.0109**. Production champion md5 is **unchanged**.

Table 1b. Multi-seed mean **test** per-class F1 (n=5; means of seed-wise class F1). Support from seed-42 test split shown for scale.

Class	Mean test F1	Test support (s42)
DDoS	0.9838	385309
DoS	0.9813	330112
Normal	0.9292	107
Reconnaissance	0.9958	18163
Theft	1.0000	14
Macro (mean±std)	0.9780±0.0033	—

Min-class F1 mean **0.9292** coincides with mean Normal F1 under this split. Perfect Theft F1 is real on this test set but must be read with support **14**.

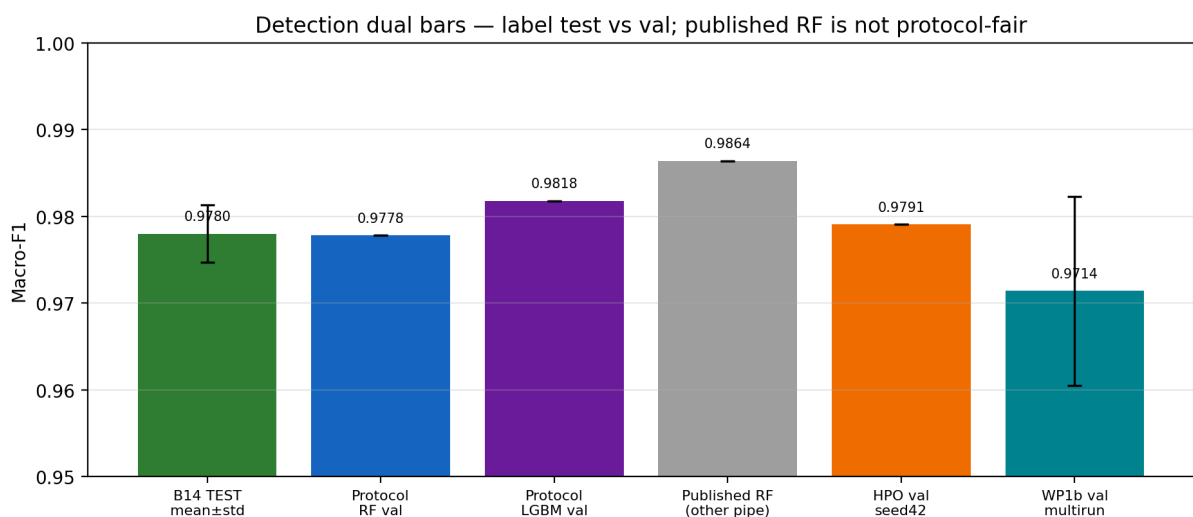


Figure 3. Dual accuracy bars: sealed multi-seed test versus protocol classical val versus published RF (other pipeline).

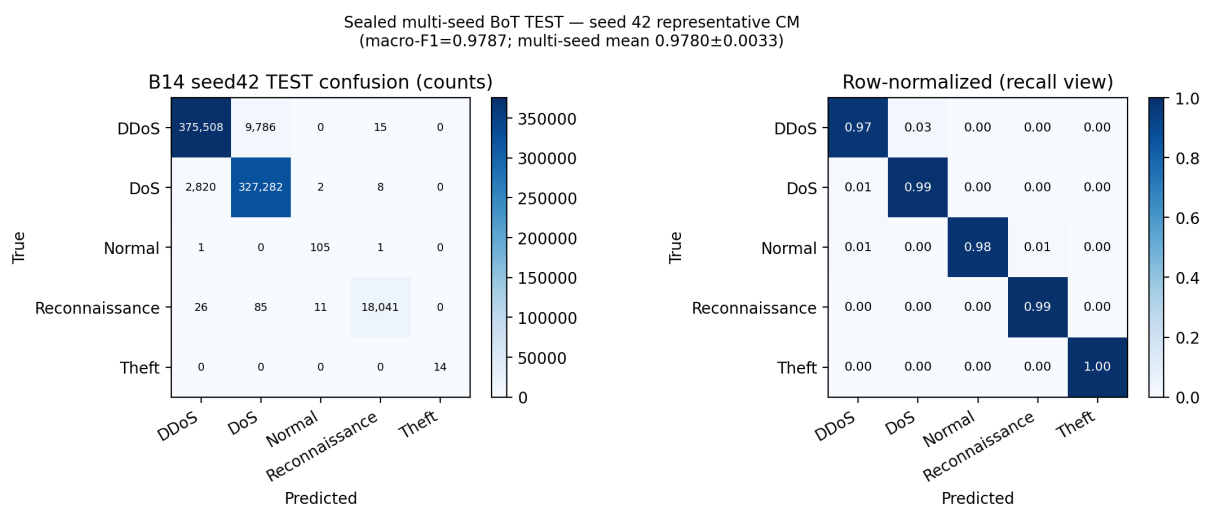


Figure 4. Representative sealed-test confusion matrix (seed 42; test macro-F1 0.9787). Multi-seed mean remains **0.9780±0.0033**.

5.2 Protocol-fair classical baselines (val)

Table 2. Classical baselines under botiot_v1 (val; test sealed during selection).

Model	val macro-F1	Notes
LGBM (balanced)	0.9818	Protocol pure-F1 ceiling
RF	0.9778	Protocol-fair
XGB	0.9762	
LR	0.5231	
LinearSVC	0.4268	Weak under extreme imbalance
Published RF	0.9864	Different pipeline — dual bar only

RQ1 answer: CAD-CBA sealed test **0.9780±0.0033** is near protocol RF and below protocol LGBM pure F1. The paper’s primary contribution is therefore **not** “we beat every classical model on F1 alone.”

5.3 Neural baselines (val, equal budget)

Table 3. Neural baselines under equal fixed CE budget (seed42).

Model	val macro-F1
G11 CNN–BiLSTM	0.9493
G6 MLP	0.9285
G10 CNN–LSTM	0.8159
G8 LSTM	0.8099
G9 BiLSTM	0.8058
G7 1D-CNN	0.6221
G12 Transformer	0.5808

Transformer is **not** a free win under this budget. G11 matches ablation A3.

5.4 Ablations and package composition

Table 4. Ablation ladder (seed42, val).

Row	val macro-F1	Role
A7 full CAD-CBA-v1	0.9699	Package top
A3 cnn_bilstm CE	0.9493	Strong backbone
A6 +ens KD	0.9346	KD lift vs A5
A5 attn+focal	0.8684	Focal helps vs A4
A2 bilstm	0.8058	
A4 attn+CE	0.7378	Attention not free
A1 cnn only	0.6221	Weak

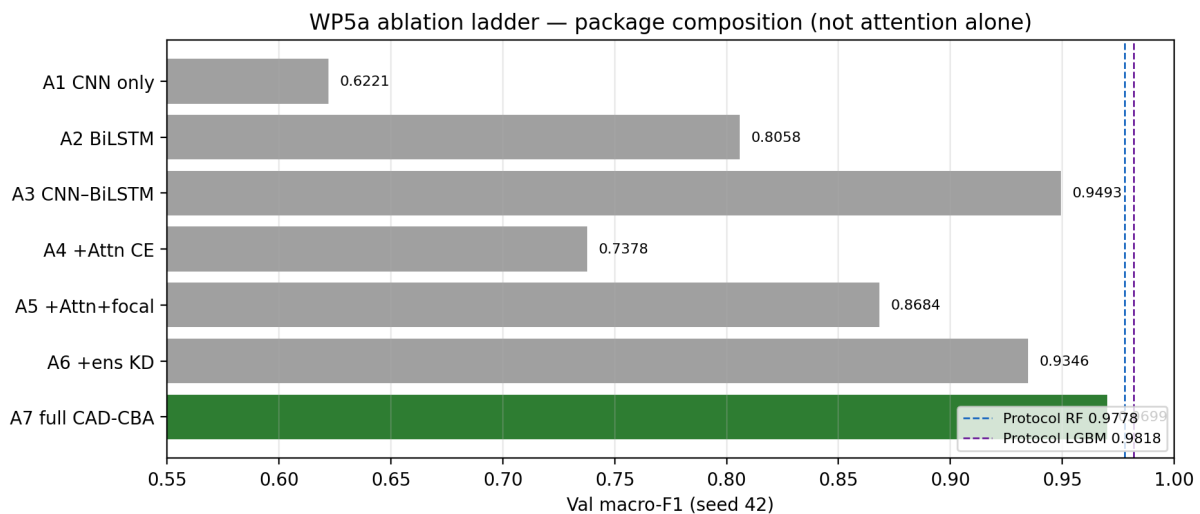


Figure 5. Ablation ladder: full package (A7) wins; attention+CE (A4) underperforms plain CNN–BiLSTM (A3).

Novelty (locked). CAD-CBA-v1 is **not** “just a standard CNN–BiLSTM.” Novelty is **composition under a frozen protocol**: focal training, ensemble tree KD, Optuna-selected train HPs, and a systems stack (export fidelity + Option A

CUDA + multi-session energy/latency). Ablations show the package path wins while attention alone is not a free gain.

5.5 Teachers / KD

Table 5. Stage-A KD student val macro-F1 ($\alpha=0.6$, $T=10$, $\gamma=2$, seed42).

Teacher	Student	Decision
ensemble	0.9401	Selected for package
RF	0.9346	Documented fallback
none	0.9326	hard-label control
XGB	0.9270	strong teacher, weaker student
LGBM	0.8829	weak path
Neural G11 (E6)	0.8513	rejected

Do not mix Stage-A KD scores with Stage-B fine-tune multirun means.

5.6 Loss / sampler / thresholds (imbalance)

Experiment	Result	Package decision
CE vs focal vs focal_cb vs logit_adj	focal 0.9780 best; CB 0.9121; logit 0.9225	focal selected
Val thresholds on focal	all variants = argmax 0.9780	keep argmax
Stratified inv-freq vs shuffle	0.9209 vs 0.9791	keep shuffle

5.7 Hyperparameter search (Optuna)

Stage A: 20 TPE trials (subsampling train, short epochs); Stage B: full-train refine of top-3. Source: benchmarks/results/hpo/summary.json.

Table 5b. Stage-B full-train refine ranking (seed42). Stage-A rank $\neq$ Stage-B rank.

Stage-A rank	Trial	Stage-A val	Full-train val	Min-cls	Theft	Package
2	8	0.9787	0.9791	0.9351	1.0000	Selected (hpo_best.yaml)
1	11	0.9787	0.9721	0.9014	1.0000	not selected
3	13	0.9786	0.8656	0.5000	0.5000	unstable under full data

Winner parameters (exact) are listed in Appendix A. Multi-seed HPO confirm (original distill + hpo_best) yields val mean **0.9689 $\pm$ 0.0145** ($n=5$); seed42 reproduces **0.9791**. The multi-seed mean does **not** beat WP1b multirun **0.9714 $\pm$ 0.0109** — train HPs remain locked, without a mean-win claim over the prior multirun.

5.8 Multi-objective Pareto

A priori composite ranking places **G6 MLP score 0.9056 @ 4.33 μ s/sample** with F1 **0.9285** first on the efficiency-weighted ranking. F1-oriented points include A7 (**0.9699 @ ~26 μ s**) and classical LGBM/RF references. Figures: figures/fig_pareto_f1_latency.png, figures/fig_pareto_f1_params.png.

Interpretation. The multi-objective story allows classical models to keep pure-F1 leadership while neural models own deployable size/latency/GPU paths — matching the contribution bar that detection need not be the sole headline.

5.9 Systems: latency, energy, VRAM (local multi-session)

Table 6. Local multi-session ranges ($n=5$ sessions, RTX 3050, champion frozen). Source: wp6b_local_ranges/summary.json.

Metric	Session-mean range	Mean $\pm$ std	CV%	95% CI
Energy mJ/flow @bs=128	0.920–0.943	0.933 $\pm$ 0.010	1.05	[0.920, 0.945]
PT μ s/sample @bs=256	24.15–25.68	24.90 $\pm$ 0.55	2.22	[24.21, 25.59]
CUDA derived pipeline μ s	565.2–570.3	567.4	0.34	—
CUDA block3 FP16 μ s	503.2–508.5	505.6	0.44	—
Peak alloc MiB	322.2	—	—	global max

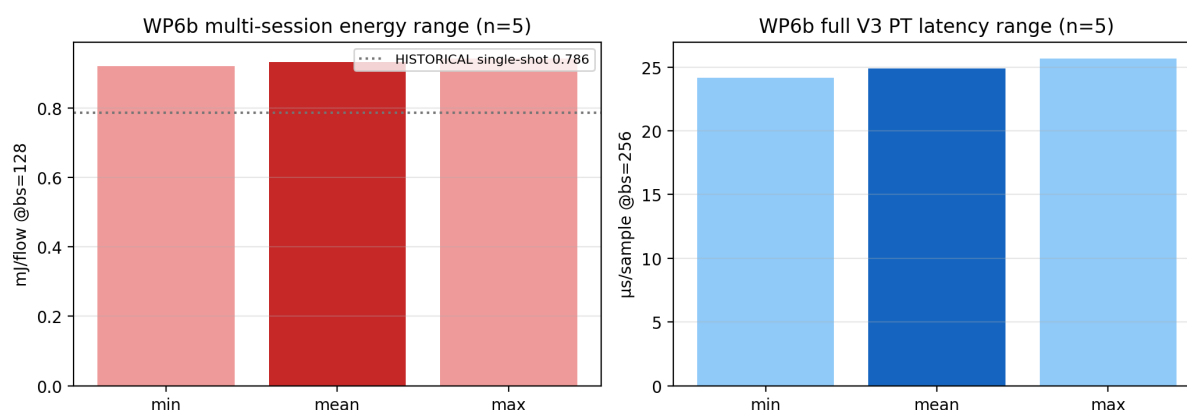


Figure 6. Local multi-session energy and PT ranges. Do **not** mix WP6b mean energy **0.933** with historical single-shot **0.786** mJ/flow (energy_table/, labeled HISTORICAL). CUDA pipeline is Option A **block sum**, not full V3 parity.

Warm-up protocol: **50** discarded sync forwards. Batch-size sensitivity: $bs \in \{1, 8, 32, 64, 128, 256, 512, 1024\}$ multi-session tables in systems_i8_h3/.

5.10 CUDA Option A and numerical fidelity

Export path: **bit-identical** real-weight blocks (max abs error 0). CUDA self-checks: **all PASS**. FP16 block3 uses documented looser tolerance. These establish kernel correctness for operation-matched blocks; they do **not** license a full-pipeline Custom CUDA versus full V3 speedup claim.

5.11 Explainability (scoped)

Table 7. XAI suite on BoT val, champion frozen (xai/summary.json).

Metric	Value	Scope
Dispatch p99 overhead	16.60 μs	Dispatch only
Occlusion rank Spearman	0.9636	Consistency
Faithfulness top-3 mass	0.5109	Proxy only
Structured usefulness (auto rubric)	1.0	Templates; not a human SOC study
Free-form feature-mention	0.333	Weak
Generation mean	~7400 ms	Never conflate with dispatch

Policy: drop a full “explainable IDS” title/abstract claim; keep dispatch + structured evidence.

5.12 Multi-dataset: ToN-IoT honesty (13-feat)

Table 8. CAD-CBA-v1 recipe transfer on processed_toniot (13 features).

Model	val macro-F1	test macro-F1
CAD-CBA-v1 (KD selected)	0.8080	0.8110
RF same-split	0.9400	0.9393

This is **not** the historical clean 26-feat CNN result (**0.9526**). The large RF gap is disclosed; this is recipe transfer, not BoT weight transfer.

5.13 Cross-GPU multi-day (TBD)

Cell	Status
V100S multi-day mean/median/std/CV/CI	TBD (DICC)
A100 multi-day	TBD (DICC)
Same-GPU B3 CUDA vs matching PT on cluster	TBD (DICC)

Local WP6b ranges must **not** be generalised to cluster portability.

6. Discussion

6.1 What the sealed test allows us to say

Multi-seed test macro-F1 **0.9780±0.0033** with Theft **1.0** is a strong, protocol-honest detection result. It is **near** protocol RF and **below** protocol LGBM on pure F1. That forces the paper's lead narrative onto **multi-objective validity** and **systems honesty**, which is exactly the contribution bar from external feedback.

6.2 Package novelty versus single-module novelty

Attention alone can hurt (A4). Multi-scale and gated variants fail bounded probes. SupCon and ASL collapse under budget. The scientific object is therefore the **evaluated package** (focal + ensemble KD + HPO + V3 + systems discipline), not a single new layer sold without measurement.

6.3 Dual bars and dual energy numbers

Always label:

- Protocol val versus sealed **test**.
- Protocol RF/LGBM versus published RF **0.9864**.
- WP6b multi-session energy **0.920–0.943** versus historical single-shot **0.786**.

6.4 Negative results as evidence

Failed variants remain in benchmarks/results/ and in this manuscript's tables. They are part of the contribution (what not to add).

6.5 Implications for deployment

For edge GPU deployments that already prefer a neural path, CAD-CBA-v1 offers near-RF multi-seed test quality, strong Theft recognition under protocol, modest peak VRAM, and quantified multi-session energy/latency ranges. Operators who optimise pure tabular F1 alone may still prefer LightGBM under this protocol. Cross-platform SLAs require DICC multi-day evidence before any portability statement.

7. Threats to validity

- **Val versus test.** Most HPO/ablation/baseline numbers are **val-only**. Sealed multi-seed **test** is separate (B14).
- **Dual accuracy bars.** Protocol RF/LGBM ≠ published RF 0.9864 pipeline.
- **Single-seed ladders.** A1–A7 and G6–G12 are seed **42** / fixed budget — multi-seed stability is B14 / multiruns.
- **Local ≠ portable.** WP6b is **RTX 3050 laptop only**.
- **Option A construct validity.** Per-block CUDA latency ≠ “full model Custom CUDA versus full V3.”
- **Energy construct.** WP6b multi-session range is primary; 0.786 is single-shot historical.
- **XAI conclusion validity.** No human SOC study; free-form LLM weak; structured templates use an automatic rubric.
- **ToN external validity.** 13-feat processed recipe transfer; large RF gap.
- **Theft support.** Test Theft n=14 — perfect Theft F1 is real under this test set but limited support must be stated.
- **Historical text blocks.** Older draft blocks may mix pre-protocol numbers; **protocol-era claims** and this manuscript override for CAD-CBA-v1.

8. Reproducibility

```
Protocol: botiot_v1 (scripts/protocol/)
Method: CAD-CBA-v1 (docs/execution_plan/METHOD_PACKAGE_DECISION.md)
Train HPS: config/hpo_best.yaml
Freeze: docs/execution_plan/FINAL_CONFIG_FREEZE_CARD.md
Champion: model/best_model_botiot_twostage.pth
md5 80a90f7cc210276300eaa90173a5a385
Sealed test: benchmarks/results/sealed_test/
Systems: benchmarks/results/wp6b_local_ranges/
Claims: docs/execution_plan/CLAIMS_REGISTRY.md
Verify: PYTHONPATH=. python3 scripts/verify_claims.py
PDF build: PYTHONPATH=. python3 scripts/build_manuscript_pdf.py
Option A: per-block CUDA only; no full-pipeline parity claims
DICC: not included until dedicated session SUCCESS tree exists
Spine: docs/execution_plan/WP9b_MANUSCRIPT_SPINE.md
This draft: docs/manuscript/CAD_CBA_v1_MANUSCRIPT.md
Figures: docs/manuscript/figures/
```

All load-bearing public numbers are registered in the claims package (**59** claims at last rebuild). Rebuild and re-verify after any prose that introduces new load-bearing numbers.

8.1 Data and code availability (PI to finalise venue wording)

- Protocol scripts, configs, and claims verifier are in the project repository.
- Result JSON under benchmarks/results/ is machine-local (often gitignored); committed manifests record md5s and headlines.
- Dataset redistribution follows original BoT-IoT / ToN-IoT licences (*PI to confirm venue data statement*).

8.2 Ethics and dual-use

This work studies intrusion detection on public research datasets. Models and CUDA kernels could be misused for adversarial research; we release evaluation protocols intended for defensive measurement. No human-subject study is reported.

9. Conclusion

We presented **CAD-CBA-v1**, a class-aware distilled CNN-BiLSTM package evaluated under a sealed, protocol-fair regime. On BoT-IoT multi-seed **test**, the method reaches **0.9780±0.0033** macro-F1 with Theft **1.0**, near protocol-fair RF, while LightGBM retains the pure-F1 ceiling (**0.9818** val). Ablations credit **package composition** rather than attention alone. Local multi-session systems ranges quantify energy (**0.920–0.943** mJ/flow), latency (PT@256 **24.15–25.68** µs), and peak VRAM (**322.2** MiB) under Option A CUDA discipline. Explainability is scoped to dispatch and structured evidence. Multi-GPU portability remains future work for a dedicated DICC session. The primary claim is therefore a **valid multi-objective accuracy–efficiency evaluation package** with honest limits — not an oversold single-number victory.

Acknowledgments

[PI: funding, compute, and collaborator acknowledgments.]

Appendix A — HPO winner parameters

HP	Value
lr	5.89306076111462e-05
batch_size	1024
focal_gamma	1.9166447754858478
dropout_rate	0.14783769837532068
attention_dropout	0.21397343616689848
weight_decay	0.00019158219548093185
scheduler	cosine

Source: config/hpo_best.yaml / WP3 trial 8 full-train refine val **0.9791**.

Appendix B — Figure checklist

Figure	File	Status
Architecture	figures/fig_architecture.png	DONE
Class distribution	figures/fig_class_distribution.png	DONE
Detection dual bars	figures/fig_detection_dualBars.png	DONE
Confusion matrix (B14 s42)	figures/fig_confusion_matrix_b14_seed42.png	DONE
Ablation ladder	figures/fig_ablation_ladder.png	DONE
WP6b systems ranges	figures/fig_wp6b_systems_ranges.png	DONE
Pareto F1–latency	figures/fig_pareto_f1_latency.png	DONE (copied)
Pareto F1–params	figures/fig_pareto_f1_params.png	DONE (copied)

Appendix C — Related-work comparison (compact)

Theme	Typical prior claim	This work
Detection	Single split / test leakage risk	Sealed multi-seed test after freeze lock
Classical comparison	Different preprocess/pipeline	Protocol-fair same split + dual bar
Acceleration	Full-pipeline CUDA vs PT	Option A per-block + absolute PT ranges
Multi-GPU	Single laptop extrapolated	Explicit TBD until DICC
XAI	"Explainable IDS" in title	Metrics first; full claim dropped

Appendix D — PI venue polish checklist

Item	Status
Continuous journal-style abstract (5-part content preserved)	DONE this pass
Author / affiliation / correspondence placeholders	DONE (PI fill)
Venue target placeholder	DONE (PI select)
Sealed-test per-class mean table (from disk)	DONE Table 1b
HPO Stage-B refine sensitivity table (from disk)	DONE Table 5b
Systems table with mean $\pm$ std / CV / CI	DONE Table 6
Softened process jargon for reader-facing prose	DONE
Reproducibility + data/ethics stubs	DONE
PDF rebuild script	<code>scripts/build_manuscript_pdf.py</code>
Final journal class file / IEEE/ACM/Elsevier template	PI after venue choice
Bibliography style + BibTeX	PI after venue choice
DICC multi-GPU cells	BLOCKED until dedicated session

Local-complete draft assembled 2026-07-22 (WP9c) · PI venue polish pass 2026-07-22. Numbers only from on-disk JSON / CLAIMS_REGISTRY. No invented DICC numbers.